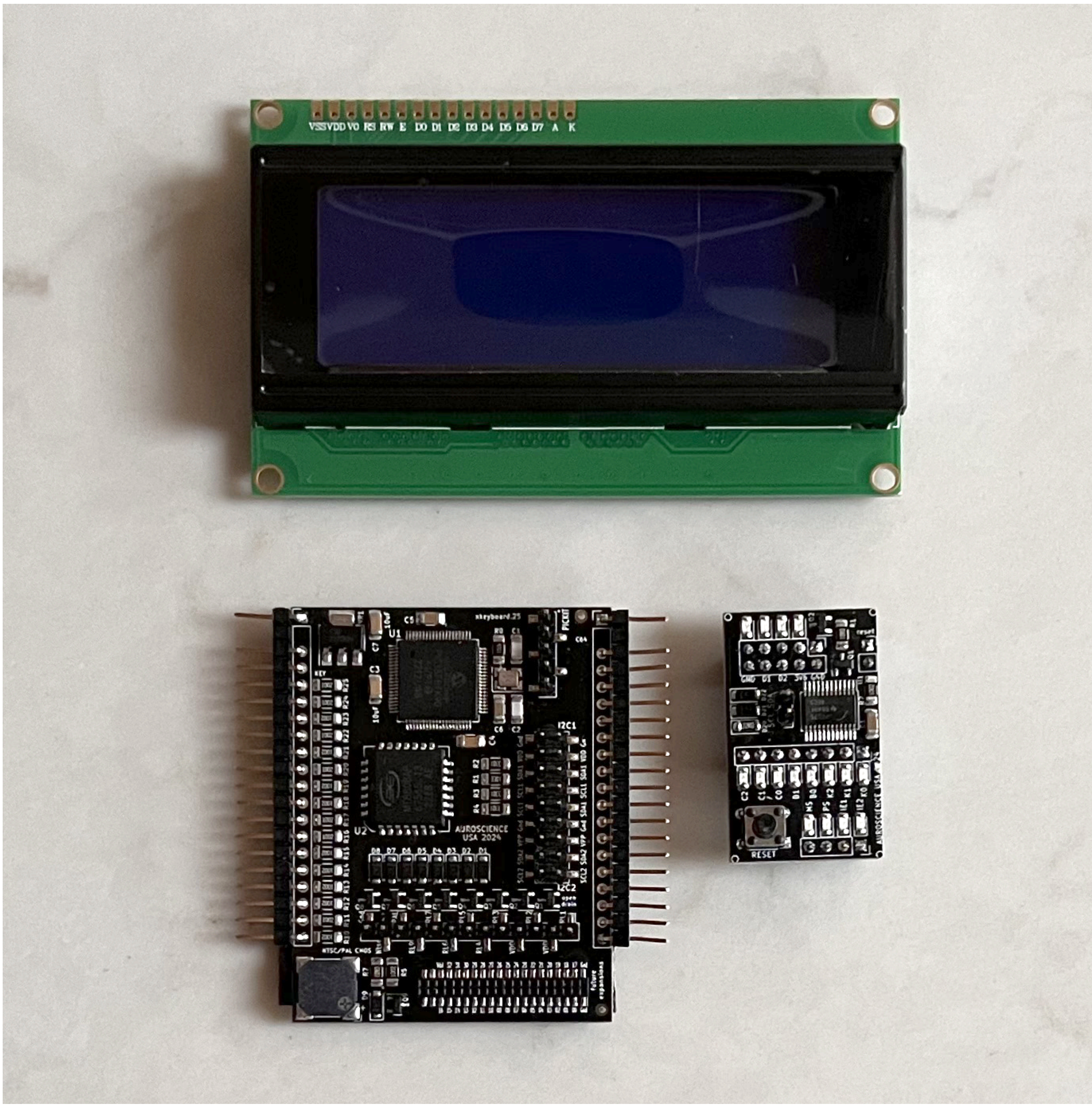
Multi Switch Module

Introducing the EVO64 Multi Switch Module (MSM)



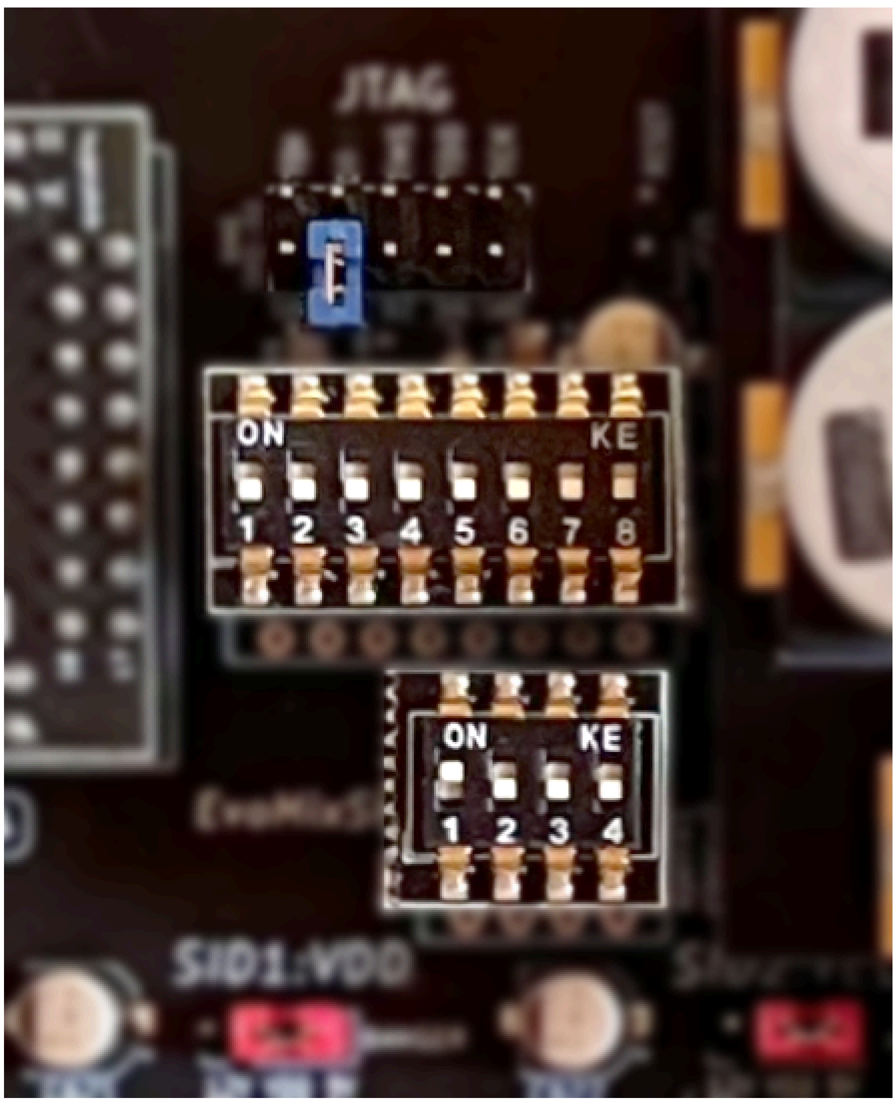
Updated: Jan 25th 2025

Background

The Multi Switch Module was developed to provide a novel and convenient solution for managing the *many* configuration options and states that the EVO64 has to offer.

Enthusiasts already know how cool it is to have onboard access to EVO's dense **multi-ROM**, containing up to 8x Kernal ROMs, 4x versions of Basic, 8x Character sets and Cartridge images. They also love the flexibility of being able to change the way their **dual-SIDs** are listening on the system's address bus, providing either dual-Mono output or true-Stereo. There's even the ability to swap which SID is primary and which address ranges the secondary SID will monitor.

However, up until now the only way to access these features was to manipulate a set of jumper headers on the EVO64 mainboard or toggle a set of dip switches, if you owned a premium parts kit.



MSM to the Rescue!

With the MSM in place, you'll hardly *ever* need to open up your system case again to manage your settings. MSM comes with three primary components that will make it much easier to make system-wide changes that will persist, even after rebooting.

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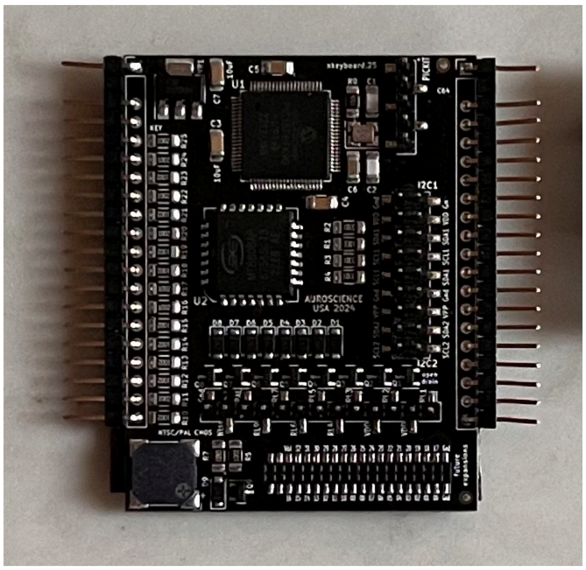
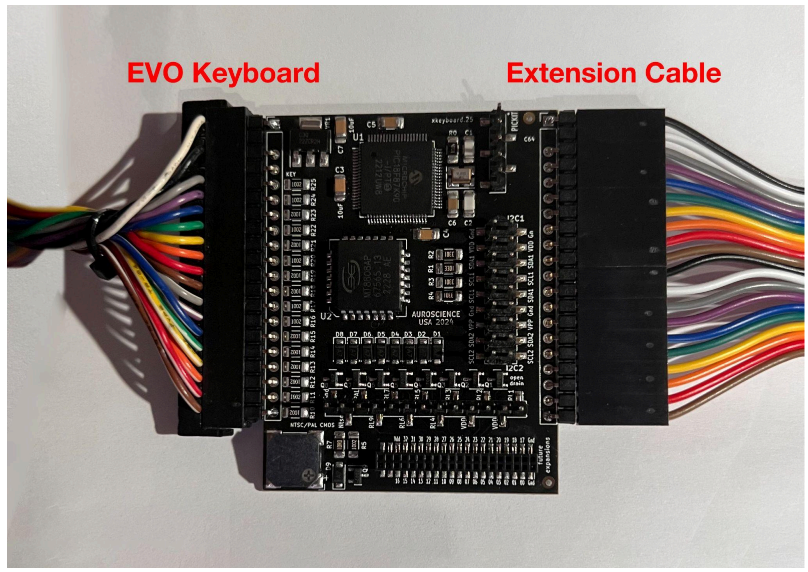
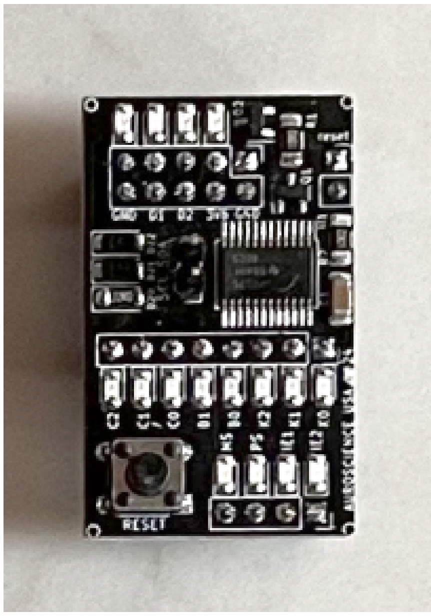


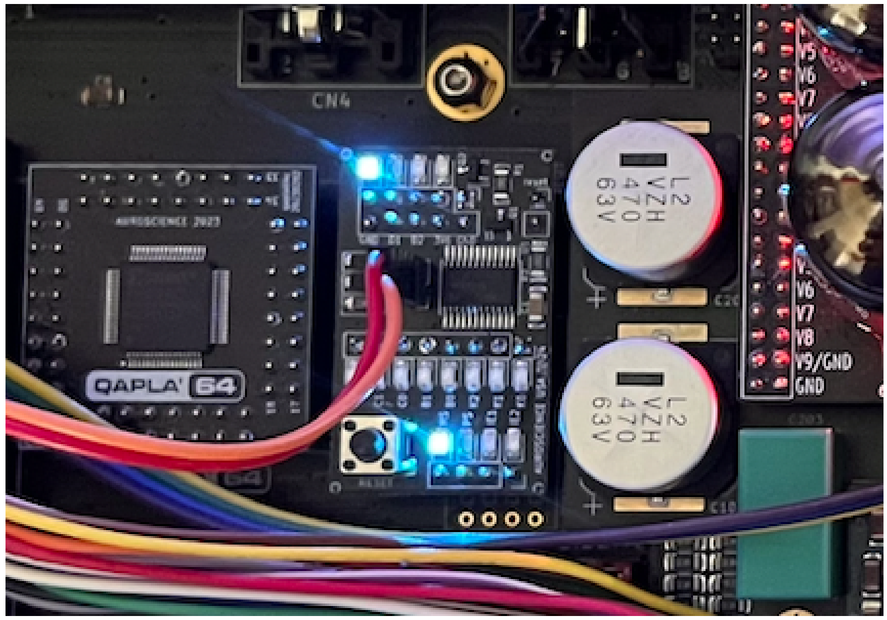


The primary module is inserted inband, between your EVO64 and its keyboard. This enables the MSM to monitor key presses until it detects a "*hot key sequence*" that tells the MSM that it’s time to issue a config change command. And, don’t worry, none of your keystrokes are *ever* collected, retained or transmitted anywhere.

The MSM gets its 5v power source directly from EVO’s keyboard connector, so there’s no need to supply any voltages separately. The MSM will also supply power to the Display Module.

ROM-SID Module





As the name implies, the ROM-SID module was developed to manage *all* of your ROM selections and SID configuration states. It replaces the need for those jumpers and dip switches. You simply install the ROM-SID module over the headers on your EVO* and use the included wires to connect it up to the MSM. *That's it!*

There's even a useful set of mini-LED lights on the module that tell you what config states are active for each of the header rows that it supports. You can get a better sense of how the LEDs relate to the configuration states by taking a look at the MSM quick reference guide:

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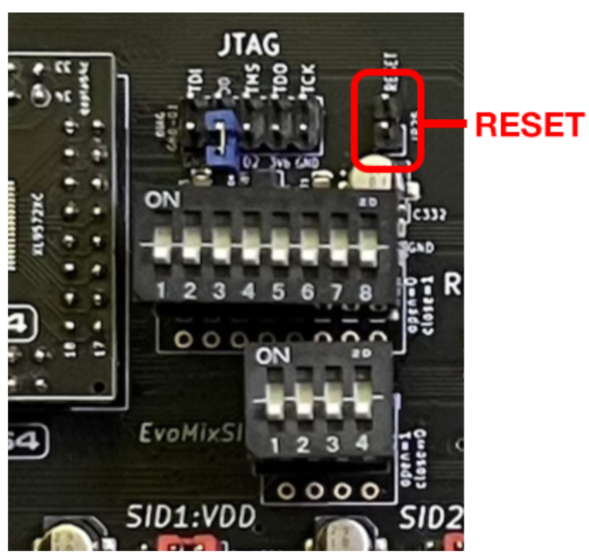
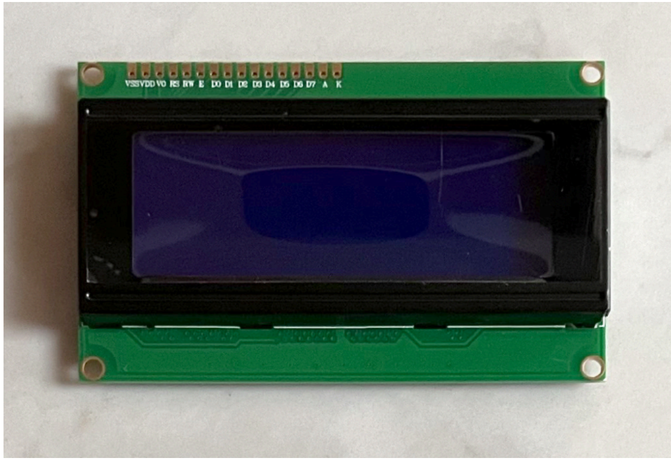
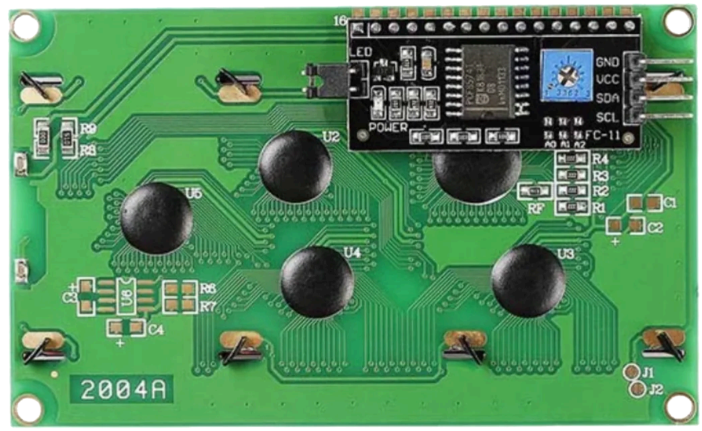
*** PLEASE NOTE:** The MSM needs to perform an automated system-reset after making any configuration changes to the ROM state (e.g., Kernals, Basic, Characters) or a Cartridge image, or a video mode (coming soon).

To facilitate the reset, your EVO system must have a 2-pin set of headers soldered into the reset line on the mainboard, nearest to where the ROM-SID module is mounted.



Config State Display Module

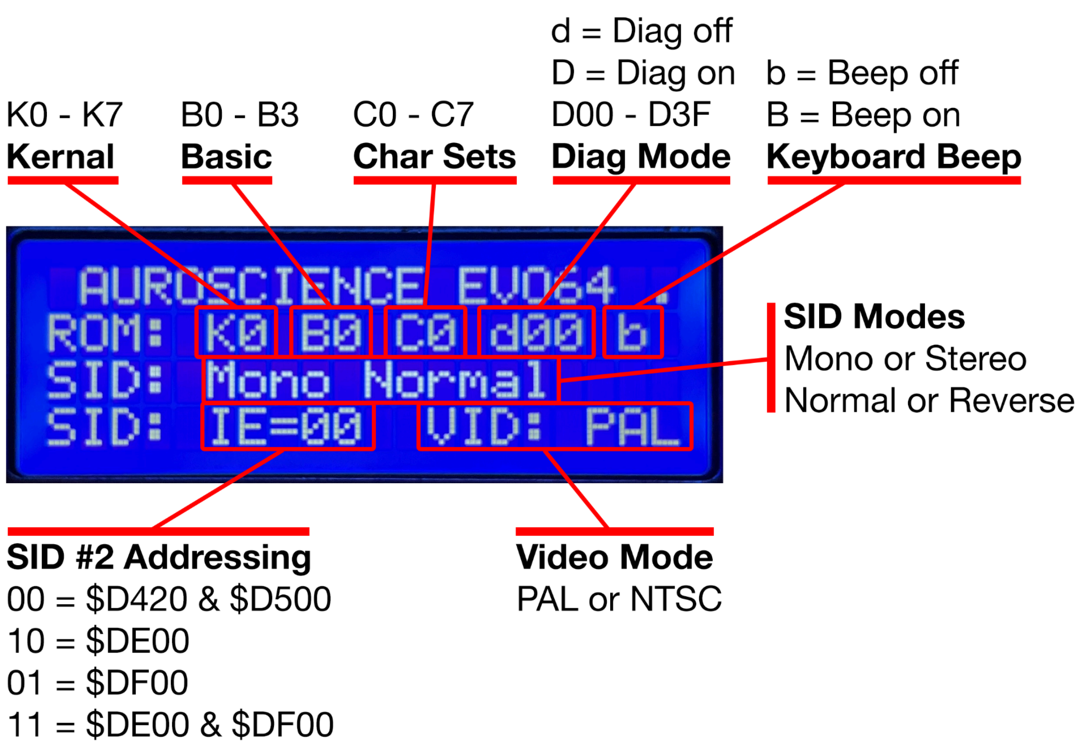




The Config State Display provides a convenient “at a glance” summary of all of the key configuration states that the MSM is managing.

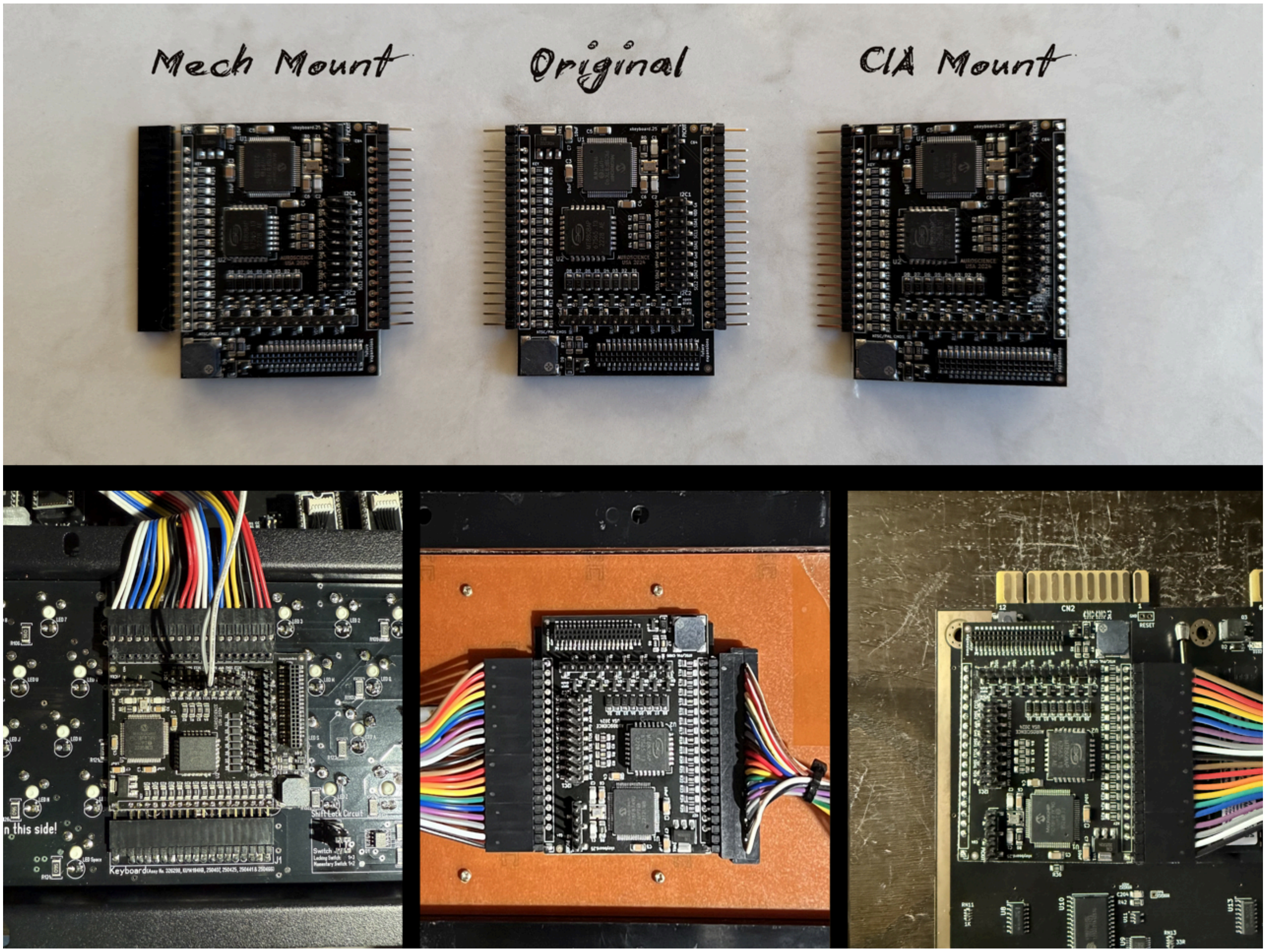
The MSM kit comes with a standard 4x20 Character display, which is the recommended display type, although there is also support for 2x16, 2x20 or 2x40 sized displays.

You’ll use the included cable to connect the display to the MSM module, using the 4 pin I2C header on the rear of the display unit.



Installing and Mounting your MSM

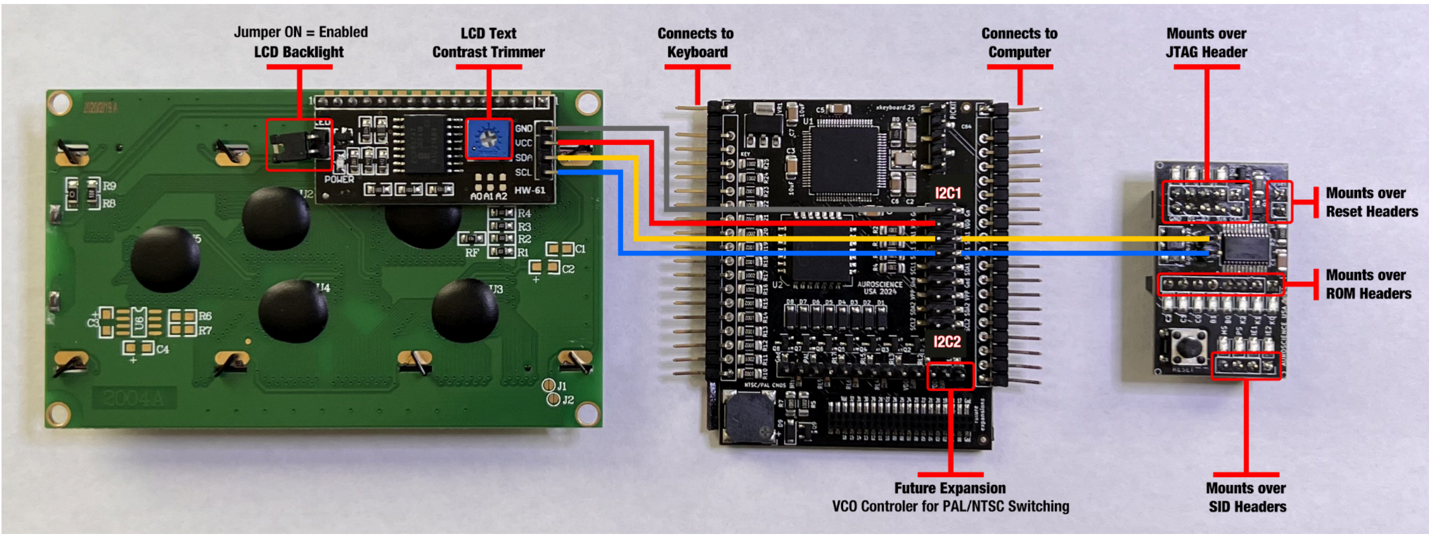
As of Jan 2025, MSM is available with three different mounting options:



MOUNTING OPTIONS:

- **Original:** Includes a 30cm extension cable, so you can place the MSM practically anywhere inside your EVO case.
- **Mech Mount:** Connects discreetly under your Mechboard64 mechanical keyboard (purchased separately) and does *not* require extension cable.
- **CIA Mount:** Plugs directly into EVO's Keyboard connector, mounting over your CIA chips. Does *not* require extension cable.

Cabling Diagram



Click [here](#) or on the image above for a full sized diagram

There's no right or wrong way to mount your MSM & Display inside of your system. It's really just a matter of preference and taste. Each style of case is going to have *very* different clearances and tolerances (e.g., Breadbin, C64c wedge, Plexilaser and the forthcoming **MODULR**), not to mention that many enthusiasts already have a variety of add-ons and upgrades mounted inside their systems.

General Use of the MSM

The MSM has been designed to be deceptively simple to use, while providing convenient access to the full scope of EVO's configuration settings.

Once your MSM is installed, it will monitor all of your keyboard activity and simply pass everything through to the system, while it waits patiently for **hot key combinations** that will activate its **Command Mode**. The circuitry is extremely low latency, so you'll never notice any delay with the MSM inband.

For convenience, we've summarized *all* of the MSM commands into a handy quick reference. Please note that there are multiple worksheets, which cover all of the ROM & Cartridge settings, as well as Audio & Video, along with a Display Module reference.

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The following sections provide more detail on the MSM command options:

SYSTEM RESET: C= + ⟵ + INST/DEL

For a convenient hot key combo to reset your system at any time, hold down the Commodore key (C= in the lower left corner of your keyboard), follow by the left Arrow key (← in the upper left corner) and the Insert / Delete key (INST DEL in the upper right corner). After a few seconds, you will see this system perform a reset. We jokingly refer to this key-combo as "**The 3-Finger Salute**". *Note that this sequence does not require the MSM to enter into command mode.*

FIRMWARE RESET: C= + ⟵ + INST/DEL + CURSOR LEFT/RIGHT

To completely clear your MSM configuration state and revert it to factory-default-settings, hold down the Commodore key (C= in the lower left corner of your keyboard), follow by the left Arrow key (← in the upper left corner), the Insert / Delete key (INST/DEL in the upper right corner) and the Left/Right Cursor key (CRSR ← / →). You will need to hold these keys longer than any of the other typical key sequences, but rest assured, after a few seconds, you will hear a series of rapid and audible beeps, then you will see this system perform a reset. Afterwards your MSM will be cleared of all configuration states. We jokingly refer to this key-combo as "**The 4-Finger Salute**". *Note that this sequence does not require the MSM to enter into command mode.*

COMMAND MODE: C= + RESTORE

This is the hot key sequence that you will use most often with the MSM. To enter COMMAND MODE, you simply hold down the Commodore key (C= in the lower left corner of your keyboard), while simultaneously pressing the RESTORE key for a couple seconds. You will hear an audible “BEEP!” sound from the MSM when you’ve entered into Command Mode, and you will see that the Config State Display has changed to show you which Command(s) are being entered.

Once you are in command mode, you simply type your command normally (only one at a time) and press RETURN to commit the command. If you make a typo mistake *before* pressing RETURN, you can simply press the DEL key to erase the most recent character(s) you entered, then retype them. If you change your mind and don’t actually want to send a command, just wait a couple of seconds for Command Mode to time out and you will see the Display return to its normal state. Afterwards you can continue typing normally.

KEYBOARD BEEP: beep

The MSM can be configured to produce a “BEEP!” sound whenever a key is pressed on the keyboard. This is a handy diagnostic feature if you’re troubleshooting faulty keys or cables and want to be sure that the keypress is actually being recognized and registered. To enable BEEP mode, you simply press the COMMAND MODE sequence and wait for the Display to change, indicating that the MSM has entered command mode. Then type the word beep, followed by the RETURN key.

DISPLAY MODULE LCD TYPE: lcd0-3

Your MSM can operate completely “*headless*” with no display connected, if that’s your preference. You will still hear audible beeps when you’ve pressed the COMMAND MODE sequence, and while you’re actively typing commands. However, we know that most people will want to leverage a display that shows them which configuration states are active at any given time.

The Config State Display provides a convenient “at a glance” summary of *all* of the key configuration states that the MSM is actively managing.

The base MSM kit comes with a standard 4x20 Character display, which is the recommended display type, although there is also support for 2x16, 2x20 or 2x40 sized displays.

To modify the LCD type, you simply press the COMMAND MODE sequence, followed by lcd#, where # is a single digit number between 0-3, followed by RETURN.

- lcd0 = 4 rows x 20 character display (default)
- lcd1 = 2 rows x 16 characters
- lcd2 = 2 rows x 20 characters

- lcd3 = 2 rows x 40 characters

SAVING & LOADING A COMPLETE SYSTEM STATE: SAVE1-8 or LOAD1-8

The MSM offers the ability to *save* and *recover* up to 8 different “complete system states”. This includes all of the ROM, SID and Video mode selections you’ve made.

This provides a super-convenient way to load multiple configuration changes to your system at the same, using a single hot key sequence.

To take advantage of the saved system state slots, you first need to set up all the configuration states for your EVO system, *one at a time*, using your MSM. In other words, you will set up each of your desired ROMs, Character Sets, SID config states, video modes, etc., using the steps provided in this guide or in the online quick reference.

Once you have your system fully dialed in, with all the configuration details you require, you simply press the COMMAND MODE sequence, followed by save#, where # is a single digit number between 1-8, then press RETURN. You can overwrite a saved slot easily, by simply repeating this action sequence.

To revert back to a saved system state at any time, you simply press the COMMAND MODE sequence, followed by load#, where # is a single digit number between 1-8, then press RETURN.

Managing ROM Settings & Cartridge Images

For convenience, the online EVO EPROM memory map reference has been updated to include the MSM command references, for *all* of the ROM and cartridge image slot states:

EVO64 EPROM Memory Map

The next section provides MSM command detail for ROMs & Cartridge Management

KERNAL ROM SELECTION: K0-7

EVO's multi-ROM provides support for up to 8x Kernal ROM images. This gives you quick access to the many cool tools, utilities, fast loaders and other conveniences that have been implemented as Kernal ROM upgrades for the C64 over the years. If your multi-ROM EPROM is already configured with multiple Kernal ROM images, you can access them by simply pressing the COMMAND MODE sequence, followed by K#, where # represents a single digit number between 0-7, then press RETURN.

BASIC ROM SELECTION: B0-3

EVO's multi-ROM provides support for up to 4x Basic ROM images. To access them, you simply press the COMMAND MODE sequence, followed by B#, where # represents a single digit number between 0-3, then press RETURN.

CHARACTER SET SELECTION: C0-7

You can also access any of the 8x Character Sets saved to your multi-ROM. To activate them, you simply press the COMMAND MODE sequence, followed by C#, where # represents a single digit number between 0-7, then press RETURN.

ENABLING CARTRIDGE / DIAGNOSTICS MODE: D00-3F

Many of the EPROMs that EVO supports for its multi-ROM have the capacity to store Cartridge images (e.g., CRT / BIN files), in addition to the system ROMs. This is a super fun way to provide instant access to onboard diagnostics, utilities, games and music making tools. Activating Cartridge / Diagnostic mode is as simple as pressing the COMMAND MODE sequence, followed by D##, where ## is a two digit value between 00 and 3F, followed by RETURN. *Please note that this is a hex / base-16 number range.*

DISABLING CARTRIDGE / DIAGNOSTIC MODE: diag

When you're done using diagnostics and having fun with your cartridges, you can use the following sequence to return to normal system ROM-based operations. Simple press the COMMAND MODE sequence, followed by diag, then press RETURN.

Here's a quick video demo:

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Managing Audio & Video Settings

For convenience, we've summarized *all* of the MSM commands into a handy quick reference. Please note that there are multiple worksheets, which cover all of the ROM & Cartridge settings, as well as Audio & Video, along with a Display Module reference.

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SID DUAL MONO MODE: SM

Placing your SIDs into dual-Mono mode is a great way to achieve a pseudo-Stereo-like effect, where two SIDs are both listening on the default $D400 address range and playing the same notes. Even though both chips are doing the same thing, the subtle differences and wear characteristics of each individual SID chip will reveal subtleties in their sound character. To activate dual-Mono mode, simply press the COMMAND MODE sequence, followed by SM and RETURN

SID TRUE STEREO MODE: SS

Placing your SIDs into true-Stereo mode is a great way to enjoy the many amazing dual-SID (e.g., 2SID) tracks that have been made for the C64. In this mode, your primary SID will continue to listen on address $D400, while your secondary SID will begin listening on both $D420 and $D500. These are the most widely used addresses for dual-SID stereo music. Although, there are a handful of tracks that require the secondary SID to listen on different addresses (see the SIE settings below). To activate true-Stereo mode, simply press the COMMAND MODE sequence, followed by SS and RETURN.

SID NORMAL / REVERSE MODES: SN / SR

EVO lets you determine which of your two SIDs will act at the primary, which means it will listen on address $D400. When in SID NORMAL MODE, the SID chip placed in the SID1 socket will be primary and SID2 will be secondary. To activate this mode, simply enter the COMMAND MODE sequence, followed by SN and RETURN. To make SID2 act at the primary, you would simply enter the SR instead, followed by RETURN.

SID EXTENDED ADDRESS MODES: SIE1-2

When you have true-Stereo mode activated you have the option to enable extended address modes for the secondary SID. There are only a few dual-SID tracks and software packages that expect the secondary SID to be available on these addresses, so they're disabled by default. When these extended address modes are activated, there's also a risk that they could interfere with some cartridge-based fast loaders, like RetroReply and others. However, it's great to know that these extended addressing options are available if and when you need them. To activate extended addressing, you simply enter the COMMAND MODE sequence, followed by SIE1 to enable address $DE00 *or* you could enter SIE2 to enable $DF00, followed by the RETURN key. Only enter one of these extended addressing commands at a time. On the extremely rare occasion when you need to activate both extended addresses, you can execute the full hot key sequence twice..

VICII VIDEO MODE: video

The MSM also has the ability to manage which crystal oscillator EVO will use to control its video mode timing. This enables you to switch between PAL and NTSC, as long as you have all the necessary add-ons. To toggle between video modes, you simply enter the COMMAND MODE sequence, followed by video and RETURN.

NOTES:

- At the time of this writing, the standard MSM kit does *not* come with a **VICII oscillator control module**.
- In addition to the VCO module, you will also need either a Kawari or a **dual-VICII swapping module** in order to be able to toggle between PAL and NTSC, without first changing your VICII chip.
- Enabling Kawari to support dynamic video mode changes requires a 2 pin header to be soldered to the board. Details are beyond the scope of this document.
- After changing a video mode, Kawari will require a complete power cycle, using the physical power switch to turn your EVO OFF, then back ON.

Here's a sneak-peak of a prototype VICII Oscillator module in action, with a Kawari

https://www.youtube.com/watch?v=ARL40zvhMDc


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